User Manual — INTA ForestMap

Forest Geospatial Analysis Platform for LiDAR and photogrammetric point clouds

Authors: Dr. Eduardo Hildt - Ing. Martín Galarza - Dr. Abel Scherf

Institution: INTA EEA Montecarlo — Forest Group

Contact: hildt.eduardo@inta.gob.ar

Version: 2026.9 — Last updated: September 2026

Contents

1. Overview 2
    Required input data 2
    Tool capabilities 2
2. System requirements 4
    Software 4
    Recommended hardware 4
3. Step-by-step installation 5
    Step 1 — Download INTA ForestMap 5
    Step 2 — Install R dependencies 5
    Step 3 — Run INTA ForestMap 5
4. Interface language 7
5. Tutorial: analyzing a point cloud 8
    5.1 Step 1 — Project configuration 8
    5.2 Step 2 — Loading and classifying the cloud 9
    5.3 Step 3 — Generating digital models 10
    5.4 Step 4 — Individual tree detection 11
    5.5 Step 5 — Exporting results 12
6. Use on Linux 14
7. Bibliographic references 15

---

1. Overview

INTA ForestMap is an interactive application for analyzing photogrammetric and LiDAR point clouds in productive forest landscapes, developed by researchers from the EEA Montecarlo and the AER Tartagal, with support from the INTA Forest Program and Drone Network. It is built entirely on the lidR ecosystem (Roussel et al., 2020), the world-reference package for processing airborne LiDAR data in forestry applications.

The tool presents the point cloud analysis workflow in an accessible graphical interface that requires no programming knowledge. The user loads their data, configures the analysis parameters with visual controls, runs the processing, and obtains all products in a results folder, together with an automatic descriptive report in HTML and PDF.

The interface is available in **7 languages** (Spanish, English, Portuguese, German, French, Italian, and Swedish), and all file names, folder names, and report paths are automatically translated into the selected language.

Required input data

File	Format	Description
LiDAR or photogrammetric point cloud	.laz / .las	File containing the x, y, z coordinates of points surveyed in a scene through a LiDAR scan or an aerophotogrammetric survey processed with photogrammetry software.
Area of interest	.shp, .kml, .geoJSON	Vector file containing the boundaries of one or several stands (polygons). Describes the boundaries of the area to be analyzed.

Tool capabilities

Function	Description
Clipping and subsampling	Delimits the analysis area using a polygon indicated by the user, clips the point cloud, and reduces the point density down to a value indicated by the user to speed up subsequent processes. The processing log reports the resulting point density (pts/m^2) both after subsampling and after noise removal.

Function	Description
Terrain classification	Separates ground and vegetation points using the CSF algorithm. This algorithm simulates a virtual mesh with a given resolution and flexibility, interpolating points where the terrain was not captured by the survey. This way it classifies the point cloud into two classes: Ground points and points above ground level (vegetation, structures, etc.).
Digital model generation	From the points classified as ground, the tool generates a Digital Elevation Model (DEM) and contour lines in vector format. From the points classified as vegetation, the tool generates a Canopy Height Model (CHM), describing the height of all vegetation relative to ground level.
Individual tree detection and canopy coverage calculation	Identifies tree tops using the LMF algorithm, searching for the highest points within a moving search window with a customizable diameter. Using a defined cutoff threshold, the area covered by canopy is mapped and the existing canopy coverage percentage is calculated.
Descriptive report	Automatically generates an HTML and PDF report with metrics, statistics, and figures describing the parameters used in the analysis and the main results obtained. File and folder paths detailed in the report are translated into the interface's active language.
Configurable export	Allows choosing whether vector layers are saved in SHP or GeoJSON format. Products are organized into a project folder named by the user.

2. System requirements

Software

Component	Minimum version	Download
R	4.0 or higher	https://cloud.r-project.org/
RStudio	Optional (if the user wants to explore other analyses with R)	https://posit.co/download/rstudio-desktop/

Note: RStudio is not required. The tool works with a base R installation only: the `install.R` and `run.R` scripts (executed by the `.cmd` / `.sh` files) handle dependency installation and launching the application without needing an IDE. RStudio remains optionally useful to facilitate error diagnostics.

Recommended hardware

Component	Recommendation
RAM	16 GB or more (for clouds with more than 10 million points)
Processor	4 or more cores (the tool takes advantage of parallel processing)
Free storage	50 GB or more

For low-density point clouds or small areas (< 5 ha), 8 GB of RAM is sufficient.

Operating system: INTA ForestMap works on both **Windows** and **Linux**. File and folder selection dialogs automatically adapt to the operating system (see section 6).

3. Step-by-step installation

Installation consists of three steps: downloading the program, installing dependencies, and running it for the first time.

Step 1 — Download INTA ForestMap

1. Open your browser and go to: https://github.com/eduardohildt/ForestMAP
2. Click the green "Code" button (top-right corner of the repository)
3. Select "Download ZIP" from the drop-down menu

The downloaded file will be named `ForestMAP-master.zip`.

4. Unzip the file to an accessible location with no special characters in the path, for example:
   - `C:\ForestMAP\`
   - `D:\Tools\ForestMAP\`

Important: In general, avoid paths with spaces, accents, or characters such as ñ.

Step 2 — Install R dependencies

The tool requires a set of R packages that are installed automatically when running the installation script. The installer works with a **base R** installation, without requiring RStudio.

On Windows:

1. Go into the unzipped `ForestMAP-master` folder.
2. Double-click `Instalar_Dependencias.cmd`. The script is in Spanish and English, showing both languages in the console.
3. Wait for the script to finish. It may take about 5 minutes. Internet access is required.
4. Accept the installation of the Pandoc package (needed to create the PDF/HTML report).
5. If errors are observed, re-run this step until the installation succeeds.

On Linux: run `Rscript dev/install.R` from a terminal located in the project folder.

What gets installed? The required packages include: shiny, lidR, terra, sf, RCSF, plotly, ggplot2, rmarkdown, knitr, pagedown, pandoc, and others. These come from the official CRAN (R) repository or from the developers' repositories (r-universe) in case they are not available on CRAN. These packages are open source.

Step 3 — Run INTA ForestMap

On Windows:

1. Double-click the `Ejecutar_ForestMap.cmd` file. The command console will load the application automatically.
2. The application will automatically open in the web browser at `http://localhost:3838`

On Linux: run `Rscript dev/run.R` from the project folder.

From this point on, only Step 3 needs to be repeated to use the tool. Installing dependencies (Step 2) is done only once.

4. Interface language

INTA ForestMap is available in **7 languages**: Spanish (ES), English (EN), Portuguese (PT), German (DE), French (FR), Italian (IT), and Swedish (SV).

The language selector is located in the top corner of the application. When switching languages:

- The entire interface (labels, buttons, messages, notifications, and the processing log) is translated immediately.
- The **output file and folder names** generated by the tool (for example `RASTER`, `VECTOR`, `REPORT`) are translated into the active language at the time of export.
- The generated **HTML/PDF report** includes detailed file paths that are also translated into the corresponding language.

This allows the same project to be fully documented in the language the user needs for their final report, without mixing terms from different languages in the results.

5. Tutorial: analyzing a point cloud

This tutorial uses the sample data included in ForestMAP, located in the `datos_ejemplo/SSP_Algarrobos/` folder. It corresponds to a silvopastoral system with algarrobo trees (*Prosopis* sp.) surveyed by photogrammetry with a DJI Air 2S drone. The photographs were processed using Agisoft Metashape, obtaining a medium-density photogrammetric point cloud. The sample files are:

- `SSP_Agarrobos_25cm.laz` — Photogrammetric point cloud already subsampled, with a uniform spacing of 25 cm between points. It is projected in Posgar 2007 Faja 3 (EPSG: 5345)
- `SSP_Algarrobos.shp` — Shapefile of the area of interest (stand)

Follow the steps in order. The application guides the workflow from left to right through its five tabs: **Configuration** → **Data Loading** → **Digital Models** → **Trees** → **Export**.

5.1 Step 1 — Project configuration

The **Configuration** tab is the entry point. Here you define the project name, load the data files, and fill in the metadata that will appear in the final report.

Project Name (required):

1. In the "Project" block, at the top of the tab, fill in the **"Project Name"** field (for example: `SSP_Algarrobos`). This field is **required**: while it is empty, the application shows a warning message and the "Continue →" button remains disabled.
2. The entered name is used to name the results folder as `Project_<entered-name>` (for example: `Project_SSP_Algarrobos`). Inside that folder, the results subfolders will be created (RASTER, VECTOR, REPORT).
3. Invalid characters for folder names (`/ \ : * ? " < > |`) and trailing spaces or periods are automatically sanitized.

Edit project data (optional):

4. Click "Edit project data" (gear icon ⚙) to fill in additional metadata that will be included in the report:
   - **Author:** analyst's name
   - **Institution:** organization or company
   - **Email:** contact email
   - **Recipient:** who the report is addressed to

Load the point cloud:

5. In the "LiDAR or photogrammetric point cloud (LAS/LAZ)" block, click "Select LAS/LAZ file".
6. Choose the file containing the point cloud to be analyzed. Navigate to `datos_ejemplo/SSP_Algarrobos/` and select `SSP_Agarrobos_25cm.laz`
7. Verify that the "✓ Loaded" indicator appears next to the file name and that the metadata (size, projection, modification date) is displayed

Load the area of interest:

8. In the "Area of interest for analysis (SHP)" block, click "Select area of interest"
9. Select `SSP_Algarrobos.shp` in the same folder
10. Verify that "✓ Selected" appears and that the polygon area is shown in hectares

If the cloud and the shapefile have different coordinate systems, the application will show a warning and automatically reproject the shapefile to the cloud's system. No additional action is required.

Select the output folder:

11. In the "Output folder for results" block, click "Select output folder"
12. Choose or create an empty folder where all products will be saved. This folder **MUST NOT BE INSIDE THE ForestMAP-master DIRECTORY**.

Continue to the next step:

13. Verify that the project name is filled in and that all three indicators show "✓" (cloud, area of interest, and output folder)
14. Click "Continue →"

5.2 Step 2 — Loading and classifying the cloud

The **Data Loading** tab performs two sequential operations: point cloud subsampling and automatic ground classification.

Configure subsampling:

In the "⚙ Point cloud subsampling" block, you control how many points per m^2 will be loaded for the analysis.

Option	Density	When to use
Low	≈ 10 pts/m^2	Quick exploration, very dense clouds, limited RAM available
Medium	≈ 25 pts/m^2	General use
High	≈ 50 pts/m^2	Better resolution of results, higher memory consumption and computation time

1. Select "Medium" (≈ 25 pts/m^2)
2. Leave the method set to "Random" (default value)

Configure ground classification (CSF):

In the "⛰ Ground classification (CSF algorithm)" block, select the type of relief of the area.

Option	Description
Flat terrain	Flat or nearly flat relief
Rolling terrain	Gentle variations — the most common
Rugged terrain	Ravines, terraces, streams

3. Select “Rolling terrain” (suitable for the sample data)

Run the loading:

4. Click “▶ Run loading and classification”
5. Wait for the processing log to display the full sequence. The log reports the **resulting point density** (in pts/m²) at two moments of the process: at the end of subsampling, and at the end of noise removal — this allows you to verify how many effective points remain available for the analysis at each stage.
6. When the “ ✅ Pre-processing complete.” notification appears, review the visualization tabs:
   - ☁️ Original cloud — cloud colored by elevation (Z)
   - 🗺️ Classified cloud — ground points in orange, vegetation in green

Use the mouse to explore the 3D cloud: click + drag to rotate, scroll wheel to zoom, right-click to pan.

5.3 Step 3 — Generating digital models

The **Digital Models** tab builds the DEM (digital elevation model), the shaded relief map (*hillshade*), the contour lines, and the CHM (canopy height model).

Configure the DEM:

In the “ ⛰️ Digital elevation model (DEM)” block:

1. **DEM resolution:** select “Medium — 0.5 m/px”. In Advanced settings, this parameter accepts values from 0.25 m/px (this floor is kept, above that of the contour lines, so as not to slow down processing on high-density clouds). For the sample data, with points spaced every 25 cm, 0.5 m is adequate.
2. **Contour line interval:** select “1 m — Normal detail”. In Advanced settings, this parameter accepts values from **0.1 m**, allowing more detailed contour lines on terrain with subtle relief.
3. **Low vegetation filtering:** leave at “Medium — 5 px” (default value).
4. **DEM and contour smoothing:** leave at “Medium — 15 px”.

Configure the CHM:

In the “ 🌿 Canopy height model (CHM)” block:

5. **CHM resolution:** select “Medium — 0.5 m/px”. In Advanced settings, this parameter accepts values from **0.05 m/px**, allowing more detail in the canopy model.

6. **Gap filling:** leave at "Medium — 0.25" (default value)

Generate the models:

7. Click "▶ Generate all models"
8. Wait for the log to indicate the completion of each step.

Explore the results:

Once the process is complete, browse through the visualization tabs:

- 🗺️ DEM — Terrain model; higher values = higher elevation areas
- 📏 Contours — Contour lines plotted over the smoothed DEM
- 🌓 Shaded relief — Hillshade: topographic visualization with simulated lighting
- 🌿 CHM — Canopy height model: grey = ground, blue = low vegetation, green = medium, yellow = high, red = emergent
- 📊 Vertical profile — Distribution of heights in the normalized cloud, with tick marks on the horizontal axis every 1 meter to facilitate reading the vegetation strata (same criterion as the tree height histogram in Step 4)

Observe the consistency of the results. If the DEM or the contours show traces of structures or low vegetation, vary the options chosen for low vegetation filtering.

5.4 Step 4 — Individual tree detection

The **Trees** tab applies the LMF (*Local Maximum Filter*) algorithm to the CHM to detect the top of each tree and calculate the stand metrics.

Configure the detection parameters:

In the " 🎄 Tree top detection" block:

1. **Search window diameter:** controls the size of the area in which each local maximum is searched. It should approximate the crown diameter of the trees in the stand.

Option	Value	When to use
Small	4 m	Young trees or small crowns
Medium	6 m	Adult trees — recommended for algarrobo trees
Large	8 m	Trees with very wide crowns

   Select "Medium — 6 m"

2. **Minimum tree height:** excludes from detection any tops below this threshold, eliminating shrubs, regeneration, or errors. Select a value appropriate to the height of the trees to be identified.
3. **Canopy coverage height cutoff:** threshold above which a CHM pixel is considered covered by canopy (it does not modify tree detection, only the canopy coverage calculation). It can be

configured taking into account the average canopy base height of the forest under study. For the example, leave at "Medium — 4 m" (default value)

Run the detection:

4. Click "▶ Detect trees"
5. Wait for the notification: " ✅ N trees detected"

Interpret the results:

The indicators at the top of the tab show:

- 🌲 Trees detected — total number of individuals
- trees/ha — stand density
- Mean height (m) — average height of the detected tops
- Max height (m) — tallest tree in the stand
- Canopy area (ha) — surface area covered by the canopy
- % Covered — percentage of the area of interest with canopy coverage

Explore the visualization tabs:

- 📍 Tree tops map — red crosses over the CHM indicating the position of each detected tree
- 📊 Distribution — histogram of the height distribution of the detected trees
- 🌳 Canopy coverage — map with areas classified as covered (canopy) and uncovered (no canopy)

Fine-tuning: if too many trees are detected (fragmentation of large crowns), increase the window diameter or the minimum height and run again. If too few are detected, reduce those parameters. The ideal point must be found where most of the tops are correctly identified.

5.5 Step 5 — Exporting results

The **Export** tab saves all products to the project folder (`Project_<name>` , inside the output folder selected in Step 1) and generates the HTML/PDF report.

The exported products are:

- 🗺️ **Raster models** — DEM, CHM, Hillshade in GeoTIFF format, saved in the `RASTER` subfolder.
- 📐 **Vector layers** — Contour lines, trees, area of interest, canopy coverage, saved in the `VECTOR` subfolder.
- 📄 **Report** — HTML and PDF with the complete analysis, saved in the `REPORT` subfolder.

Choose the vector layer format:

1. In the " 📐 Vector format" block, choose between **SHP** (Shapefile, classic format compatible with most GIS) or **GeoJSON** (plain-text, lightweight, and easy to version). This choice applies to all exported vector layers (contour lines, trees, area of interest, and canopy coverage).

Export the products:

2. Click "💾 📄 Export & generate report"
3. Follow the export log — now fully translated into the active language, including folder creation, the progress of each product type, and any errors.
4. Wait for the notification: "✅ Export complete."

Generate the HTML/PDF report:

5. Wait for the notification: "✅ HTML report generated."

Report generation may take 1–2 minutes since it includes high-resolution figures. The detailed file paths within the report (for example, the path to the `RASTER` or `VECTOR` folder) are shown translated into the interface's active language.

Access the results:

6. Click "📂 Open folder" to open the results folder directly in the file explorer

Use in QGIS: GeoTIFF and Shapefile/GeoJSON files can be loaded directly into QGIS to build a GIS project with all the analysis products. Drag the files from the file explorer into the QGIS layers panel.

6. Use on Linux

INTA ForestMap works on Linux with the same behavior as on Windows. The only difference is the internal mechanism used to display the file and folder selection dialogs:

- **On Windows**, the application uses the operating system's native dialog boxes (unchanged from previous versions).
- **On Linux** (outside of RStudio), the application automatically uses the dialog boxes from the **Tcl/Tk** package, included with the standard R installation. No additional configuration is required from the user: the tool detects the operating system and the execution environment, and transparently chooses the appropriate mechanism.
- If the application is run inside RStudio (on any operating system), RStudio's own dialogs are used.

To install and run the tool on Linux, see the alternative commands provided in section 3 (`Rscript dev/install.R` and `Rscript dev/run.R`).

7. Bibliographic references

Avery, T. E., & Burkhart, H. E. (2002). *Forest Measurements* (5th ed.). McGraw-Hill.

Roussel, J.-R., Auty, D., Coops, N. C., Tompalski, P., Goodbody, T. R. H., Meador, A. S., Bourdon, J.-F., de Boissieu, F., & Achim, A. (2020). lidR: An R package for analysis of Airborne LiDAR Data. *Remote Sensing of Environment*, 251, 112061. https://doi.org/10.1016/j.rse.2020.112061

Roussel, J.-R., & Auty, D. (2023). *Airborne LiDAR Data Manipulation and Visualization for Forestry Applications*. R package version 4.x. https://cran.r-project.org/package=lidR

Zhang, W., Qi, J., Wan, P., Wang, H., Xie, D., Wang, X., & Yan, G. (2016). An easy-to-use airborne LiDAR data filtering method based on cloth simulation. *Remote Sensing*, 8(6), 501. https://doi.org/10.3390/rs8060501

Document prepared by INTA EEA Montecarlo. For technical support: hildt.eduardo@inta.gob.ar